

Mandelbrot Rulers as Observation-Scale Geometry

Dimensionless traces, protocol equivalence, finite-window exponents, and uncertainty

Stassis Research Program

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Status. This revision supersedes v1.0. Every physical scale is normalized by a positive reference length before entering a logarithm. Intrinsic object scales are separated from instrument resolutions. Finite-scale exponents require a declared scale window, weighting rule, residual diagnostic, and uncertainty semantics. A statistical confidence interval is forbidden unless a stochastic error model is supplied.

Abstract

Mandelbrot's coastline problem is naturally expressed as a family of scale-indexed observation protocols rather than as a single instrument-free length. For a bounded set A in a metric space, a protocol records a positive reference length ℓ_* , physical resolutions ε , a covering, divider, or dilation rule, and an error model. The dimensionless coordinate $t = \log_b(\ell_*/\varepsilon)$ and the trace $y(t) = \log_b N_\varepsilon(A)$ are invariant under coherent changes of length unit. Lower and upper box dimensions are independent of both ℓ_* and the logarithm base. Lipschitz observations cannot increase upper box dimension at matched scales, while bi-Lipschitz observations preserve it. Protocols whose counts are comparable up to fixed multiplicative and scale constants have the same asymptotic exponent, although finite-scale amplitudes and residuals may differ. For finite data, the weighted slope estimator is defined together with its window; a deterministic stability interval is distinguished from a probabilistic confidence interval. The Koch curve supplies an exact divider benchmark. Natural coastlines need not possess a unique universal exponent outside a declared finite scale regime.

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1 Scope and claim firewall

This module uses Mandelbrot’s ruler argument as a strict test of the distance-scale interface. It does not replace Hausdorff measure, Minkowski content, or geometric measure theory. It distinguishes three levels:

$$\text{set and metric} \longrightarrow \text{scale-indexed protocol} \longrightarrow \begin{matrix} \text{asymptotic or} \\ \text{finite-window summary} \end{matrix}$$

Boundary 1.1 (Claim boundary). Scale-dependent measurements do not imply that geometry is subjective. The set, metric, protocol, and scale are mathematical data. A log-log line over a short range is not proof of exact self-similarity. Different protocols may share an exponent while differing in prefactor, oscillation, lacunarity, bias, and uncertainty.

2 Typed scale protocol

Let (X, d) be a metric space whose distance has scale type U_L , and let $A \subseteq X$ be nonempty and bounded.

Definition 2.1 (Intrinsic descriptor). The intrinsic descriptor is

$$\xi_A = (A, d, \text{diam}_d A, \text{regularity and topology data}, \dots).$$

It contains properties of the mathematical object and metric, not the instrument resolution.

Definition 2.2 (Scale-observation protocol). A finite scale protocol is a tuple

$$\mathcal{P} = (\ell_*, b, I, \{\varepsilon_i\}_{i=1}^m, \mathcal{M}, \mathbf{A}, \vartheta, \mathcal{W}, \mathbf{E}),$$

where

- (i) $\ell_* > 0$ is a reference length with type U_L ;
- (ii) $b > 1$ is the logarithm base;
- (iii) $I = [\varepsilon_{\min}, \varepsilon_{\max}]$ is the physical scale range;

- (iv) each $\varepsilon_i \in I$ is a positive instrument resolution;
- (v) $\mathcal{M}$ is the measurement family (cover, divider, dilation, or another declared operator);
- (vi) A is the anchoring, grid, stopping, or boundary convention;
- (vii) ϑ collects nuisance settings such as grid translation, rotation, and sampling phase;
- (viii) $\mathcal{W} \subseteq \{1, \dots, m\}$ is a fit window;
- (ix) E is an exactness, deterministic-error, or stochastic error model.

The scale coordinate is

$$t_i := \log_b \frac{\ell_*}{\varepsilon_i}.$$

Its argument is dimensionless. A bare expression $\log(1/\varepsilon)$ is prohibited when ε is a physical length.

Definition 2.3 (Scale trace). For a protocol output $Y_{\mathcal{P}}(A; \varepsilon)$, the finite trace is

$$T_{\mathcal{P}}(A) := \{(t_i, Y_{\mathcal{P}}(A; \varepsilon_i)) : 1 \leq i \leq m\}.$$

An asymptotic trace additionally specifies the directed limit $\varepsilon \downarrow 0$ and the admissible protocol family.

3 Covering trace and box dimensions

Use internal closed-ball covers:

$$N_{\varepsilon}(A, d) = \min \left\{ |S| : S \subseteq A, A \subseteq \bigcup_{s \in S} \overline{B}_d(s, \varepsilon) \right\},$$

with value $+\infty$ if no finite cover exists. For totally bounded A , the count is finite for every $\varepsilon > 0$.

Definition 3.1 (Covering entropy). For $b > 1$,

$$\mathcal{H}_{b, \varepsilon}(A, d) := \log_b N_{\varepsilon}(A, d).$$

This is a covering-complexity quantity with information unit fixed by b ; it is not thermodynamic entropy.

Proposition 3.2 (Scale monotonicity). *If $0 < \varepsilon_1 \leq \varepsilon_2$, then*

$$N_{\varepsilon_1}(A, d) \geq N_{\varepsilon_2}(A, d).$$

Consequently $t \mapsto \mathcal{H}_{b, \ell_ b^{-t}}(A, d)$ is nondecreasing.*

Proof. Every ε_1 -ball lies in the ball with the same center and radius ε_2 , so any ε_1 -cover can be enlarged without increasing its cardinality. $\square$

Definition 3.3 (Lower and upper box dimensions).

$$\underline{\dim}_B A := \liminf_{\varepsilon \downarrow 0} \frac{\log_b N_{\varepsilon}(A, d)}{\log_b(\ell_*/\varepsilon)},$$

$$\overline{\dim}_B A := \limsup_{\varepsilon \downarrow 0} \frac{\log_b N_\varepsilon(A, d)}{\log_b(\ell_*/\varepsilon)}.$$

If they agree, their common value is $\dim_B A$.

Theorem 3.4 (Reference and base independence). *The lower and upper dimensions in definition 3.3 are independent of the positive reference length ℓ_* and of the logarithm base $b > 1$.*

Proof. Changing the base multiplies numerator and denominator by the same positive constant. Replacing ℓ_* by $c\ell_*$ adds the constant $\log_b c$ to the denominator. Since $\log_b(\ell_*/\varepsilon) \rightarrow +\infty$, this constant does not alter the liminf or limsup quotient. $\square$

Remark 3.5. Reference independence is an asymptotic statement. A finite-window intercept depends on ℓ_* , while a fitted slope with an intercept does not.

4 Observation maps and dimension

Theorem 4.1 (Matched Lipschitz cover). *If $O : (X, d_X) \rightarrow (Y, d_Y)$ is L_O -Lipschitz with $L_O > 0$, then*

$$N_{L_O\varepsilon}(O(A), d_Y) \leq N_\varepsilon(A, d_X).$$

Hence

$$\overline{\dim}_B O(A) \leq \overline{\dim}_B A, \quad \underline{\dim}_B O(A) \leq \underline{\dim}_B A$$

whenever the displayed dimensions are defined by compatible reference scales.

Proof. The image of every d_X -ball of radius ε is contained in a d_Y -ball of radius $L_O\varepsilon$. Map a finite input cover into the output. For the dimension statement, take logarithms and note that the fixed scale factor L_O shifts the logarithmic denominator by a constant. $\square$

Corollary 4.2 (Bi-Lipschitz invariance). *If, for some $0 < m_O \leq L_O < \infty$,*

$$m_O d_X(x, x') \leq d_Y(Ox, Ox') \leq L_O d_X(x, x')$$

on A , then lower and upper box dimensions are preserved.

Proof. Apply theorem 4.1 to O and to the inverse $O^{-1} : O(A) \rightarrow A$, which is m_O^{-1} -Lipschitz. $\square$

Thus the exponent is an invariant of the bi-Lipschitz observation class, not of arbitrary lossy observations.

5 When different rulers have the same exponent

Finite-scale traces depend on the protocol. The following criterion separates protocol dependence from asymptotic exponent dependence.

Definition 5.1 (Exponent-equivalent protocols). Two positive count traces $N_\varepsilon^{(1)}$ and $N_\varepsilon^{(2)}$ are exponent-equivalent if there exist constants $a_1, a_2, c_1, c_2 > 0$ and $\varepsilon_0 > 0$ such that, for $0 < \varepsilon < \varepsilon_0$,

$$c_1 N_{a_1\varepsilon}^{(1)} \leq N_\varepsilon^{(2)} \leq c_2 N_{a_2\varepsilon}^{(1)}.$$

Theorem 5.2 (Protocol-comparison criterion). *Exponent-equivalent protocols have the same lower and upper logarithmic scaling exponents.*

Proof. Take logarithms. The constants c_j add bounded terms to the numerators, and the constants a_j add bounded terms to the logarithmic scale coordinates. Division by $\log(\ell_*/\varepsilon) \rightarrow +\infty$ removes all such bounded terms. Apply the two inequalities to obtain matching liminf and limsup bounds. $\square$

Remark 5.3. Theorem 5.2 does not say that the traces are equal. Prefactors, log-periodic residuals, crossovers, and finite-window biases remain protocol-dependent.

For Euclidean subsets, standard axis-aligned cube covers, ball covers, and comparable grids satisfy relations of this type, so they recover the same box exponent even though their finite counts differ.

6 Divider, covering-length, and dilation protocols

6.1 Divider or ruler walk

A ruler walk along an ordered curve requires a stopping and anchoring rule. There is no protocol-free integer $N_\varepsilon^{\text{ruler}}$. Once a rule $\mathbf{A}$ is declared, define

$$L_\varepsilon^{\text{ruler}}(A; \mathbf{A}) := \varepsilon N_\varepsilon^{\text{ruler}}(A; \mathbf{A}).$$

The product has physical dimension L .

If, for some D and dimensionless $C_{\mathbf{A}} > 0$,

$$N_\varepsilon^{\text{ruler}}(A; \mathbf{A}) = C_{\mathbf{A}} \left(\frac{\ell_*}{\varepsilon} \right)^D (1 + o(1)),$$

then the normalized measured length obeys

$$\frac{L_\varepsilon^{\text{ruler}}}{\ell_*} = C_{\mathbf{A}} \left(\frac{\varepsilon}{\ell_*} \right)^{1-D} (1 + o(1)).$$

For $D > 1$ this diverges as $\varepsilon \downarrow 0$; for $D = 1$ it can converge to a finite protocol-dependent prefactor.

Boundary 6.1 (No universal divider). For self-intersections, inlets narrower than ε , disconnected sets, or digitized curves, a greedy divider may skip different pieces depending on traversal and anchoring. The algorithm is therefore part of the observation map.

6.2 Covering-length proxy

For a curve-like set one may define the distinct proxy

$$L_\varepsilon^{\text{cov}}(A) := \varepsilon N_\varepsilon(A, d).$$

This is not automatically the arclength of a curve. It is a typed scale-dependent proxy derived from covering complexity.

6.3 Minkowski sausage

For bounded $A \subset \mathbb{R}^m$, let

$$A_\varepsilon := \{x \in \mathbb{R}^m : \text{dist}(x, A) \leq \varepsilon\}, \quad V_A(\varepsilon) := \text{Vol}_m(A_\varepsilon).$$

The normalized volume trace is

$$\bar{V}_A(\varepsilon) := \frac{V_A(\varepsilon)}{\ell_*^m}.$$

If the limit exists, a Minkowski exponent can be written dimensionally as

$$D_M = m - \lim_{\varepsilon \downarrow 0} \frac{\log_b \bar{V}_A(\varepsilon)}{\log_b(\varepsilon/\ell_*)}.$$

Equivalently, finite nonzero Minkowski content corresponds to

$$\bar{V}_A(\varepsilon) = C_M \left(\frac{\varepsilon}{\ell_*} \right)^{m-D_M} (1 + o(1)),$$

with dimensionless $C_M > 0$.

7 Finite-window exponent estimation

An asymptotic dimension is a limit. Experimental and numerical data are finite. Let

$$t_i = \log_b \frac{\ell_*}{\varepsilon_i}, \quad y_i = \log_b N_{\varepsilon_i}(A, d), \quad i \in \mathcal{W},$$

where $\mathcal{W}$ contains at least two distinct values of t_i .

Definition 7.1 (Weighted finite-window slope). For declared weights $w_i > 0$, set

$$\begin{aligned} \bar{t}_w &= \frac{\sum_{i \in \mathcal{W}} w_i t_i}{\sum_{i \in \mathcal{W}} w_i}, & \bar{y}_w &= \frac{\sum_{i \in \mathcal{W}} w_i y_i}{\sum_{i \in \mathcal{W}} w_i}, \\ \hat{D}_{\mathcal{W},w} &= \frac{\sum_{i \in \mathcal{W}} w_i (t_i - \bar{t}_w)(y_i - \bar{y}_w)}{\sum_{i \in \mathcal{W}} w_i (t_i - \bar{t}_w)^2}. \end{aligned}$$

The fitted intercept is $\hat{\alpha} = \bar{y}_w - \hat{D}_{\mathcal{W},w} \bar{t}_w$.

Theorem 7.2 (Unit, reference, and base covariance of the slope). *The slope in definition 7.1 is unchanged by:*

- (i) *coherent changes of length unit* $(\ell_*, \varepsilon_i) \mapsto (a\ell_*, a\varepsilon_i)$;
- (ii) *replacing ℓ_* by $c\ell_*$ while retaining an intercept*;
- (iii) *changing the logarithm base simultaneously in t_i and y_i .*

Proof. In (i), every ratio ℓ_*/ε_i is unchanged. In (ii), all t_i receive the same additive constant, which cancels in $t_i - \bar{t}_w$. In (iii), all t_i and y_i are multiplied by the same positive constant, so numerator and denominator receive the same squared factor. $\square$

Definition 7.3 (Residual diagnostics). Let

$$r_i = y_i - \hat{\alpha} - \hat{D}_{\mathcal{W},w} t_i.$$

A minimum deterministic report contains

$$\left(\mathcal{W}, \{\varepsilon_i\}, \{w_i\}, \hat{D}_{\mathcal{W},w}, \max_{i \in \mathcal{W}} |r_i|, \sqrt{\frac{\sum w_i r_i^2}{\sum w_i}} \right).$$

Definition 7.4 (Deterministic pairwise-slope range).

$$I_{\mathcal{W}}^{\text{pair}} := \left[\min_{\substack{i < j \\ i, j \in \mathcal{W}}} \frac{y_j - y_i}{t_j - t_i}, \max_{\substack{i < j \\ i, j \in \mathcal{W}}} \frac{y_j - y_i}{t_j - t_i} \right].$$

This interval measures finite-window slope variation. It is not a confidence interval and carries no coverage probability.

7.1 When a confidence interval is legitimate

Suppose repeated measurements or a declared stochastic model justify

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\eta}, \quad \boldsymbol{\beta} = (\alpha, D)^\top, \quad \mathbb{E}[\boldsymbol{\eta}] = 0, \quad \text{Cov}(\boldsymbol{\eta}) = \boldsymbol{\Sigma},$$

where $\mathbf{X}$ has rows $(1, t_i)$ and $\boldsymbol{\Sigma}$ is positive definite. The generalized least-squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{y}.$$

If $\boldsymbol{\eta}$ is multivariate Gaussian and $\boldsymbol{\Sigma}$ is known, then

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1},$$

and a normal-theory interval for D is available. If variance parameters are estimated, the corresponding degrees of freedom and estimator must be declared.

Boundary 7.5 (Correlated scales). Counts at nested resolutions are usually correlated because they are computed from the same set. Treating the log-counts as independent homoscedastic observations is not a default law. Without an error model, report deterministic residual and window stability only.

8 Grid phase and nuisance-parameter uncertainty

For box counts in $\mathbb{R}^m$, let ϑ encode grid translation and, when relevant, rotation. The finite count is then

$$N_{\varepsilon}^{\text{grid}}(A; \vartheta).$$

The protocol must choose one of the following:

- (a) a fixed, reproducible ϑ ;
- (b) an extremal count over a declared nuisance set;
- (c) an average or quantile under a declared probability law for ϑ .

A spread over arbitrarily chosen grid phases is a sensitivity analysis, not automatically a sampling confidence interval. Pixelization, segmentation, map projection, and boundary extraction add further protocol layers and must not be absorbed into the exponent without a record.

9 Exact benchmark: the Koch curve

Let the initial segment have length $\ell_0 > 0$. At construction stage n , the generator-aligned segment ruler has

$$\varepsilon_n = \ell_0 3^{-n}, \quad N_n^{\text{seg}} = 4^n.$$

Therefore

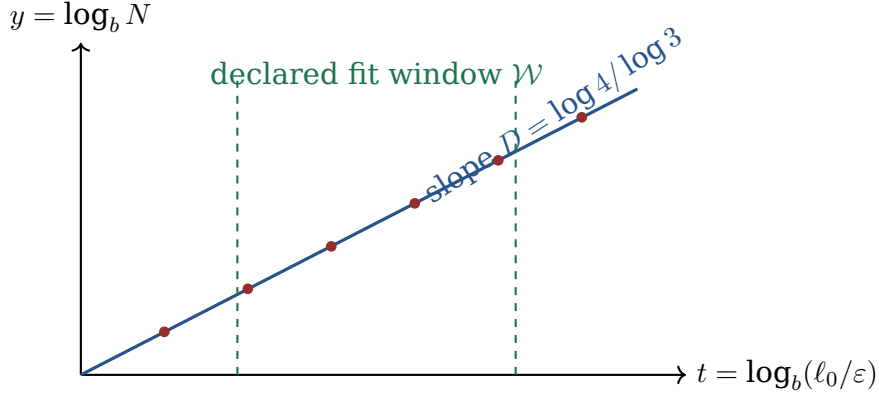
$$t_n = \log_b \frac{\ell_0}{\varepsilon_n} = n \log_b 3, \quad y_n = \log_b N_n^{\text{seg}} = n \log_b 4,$$

and every nontrivial exact stage window gives

$$\hat{D} = \frac{\log_b 4}{\log_b 3} = \frac{\log 4}{\log 3}.$$

The measured divider length satisfies

$$\frac{L_n}{\ell_0} = \frac{N_n^{\text{seg}} \varepsilon_n}{\ell_0} = \left(\frac{4}{3}\right)^n = \left(\frac{\varepsilon_n}{\ell_0}\right)^{1-D}.$$



Remark 9.1. The equality $N_n^{\text{seg}} = 4^n$ belongs to the generator-aligned divider protocol. It is not asserted as the exact count for every axis-aligned box grid. Standard covering protocols recover the same asymptotic dimension through comparison estimates.

The zero residual of this exact synthetic benchmark is not evidence that a natural coastline follows the same law.

10 Natural boundaries and finite scaling regimes

For empirical boundaries there may be lower and upper cutoffs:

$$\varepsilon_{\text{pixel}} \lesssim \varepsilon \lesssim \varepsilon_{\text{domain}}.$$

Below the first cutoff, segmentation or instrument noise dominates. Above the second, the finite domain and large-scale geometry dominate. Intermediate intervals can contain several crossovers rather than one power law.

A reproducible empirical statement has the form:

Under protocol $\mathcal{P}$, on the scale window $[\varepsilon_a, \varepsilon_b]$, the declared estimator gives $\hat{D}$, with the listed residual diagnostic and uncertainty model.

It must not be shortened to “the object has dimension $\hat{D}$ ” unless the limiting hypotheses are independently justified.

11 Connection to GO metric entropy

The metric-entropy module supplies the trace

$$\mathcal{H}_{b,\varepsilon}(A, d) = \log_b N_\varepsilon(A, d).$$

The present module adds:

- (i) the common distance-scale passport;
- (ii) the dimensionless coordinate $t = \log_b(\ell_*/\varepsilon)$;
- (iii) observation-map transformation laws;
- (iv) finite-window estimation and uncertainty semantics;
- (v) protocol-equivalence criteria.

When a differentiable interpolation is justified, an effective local exponent may be written

$$D_{\text{eff}}(t) = \frac{d\mathcal{H}(t)}{dt}.$$

For stepwise exact counts, this derivative is not canonical; a finite difference or smoothing rule is part of the estimator.

12 Normative reporting contract

For exact synthetic sets, the channel is explicitly noiseless. Temporal resolution and observation horizon are not-applicable for a static boundary; they must be supplied for an evolving boundary or time series. This module reports traces and exponent diagnostics rather than a scalar GO defect, so defect-normalizations is not-applicable until a downstream decision loss is introduced.

Field	Type	Required content
Object A and metric d	set and U_L -valued metric	domain, boundedness, total boundedness, metric convention
Intrinsic descriptor ξ_A	system data	diameter, regularity, and known cutoffs
Measurement family $\mathcal{M}$	operator	cover, divider, dilation, or explicit alternative
Noise/channel policy	kernel or exact flag	stochastic model or explicit noiseless statement
Resolution ε	U_L , positive	sampled values and ordering
Temporal resolution/horizon	time or not applicable	mandatory for time-dependent traces
Reference length ℓ_*	U_L , positive	normalization and unit context
Logarithm base b	dimensionless, $b > 1$	information unit
Anchoring A	protocol metadata	grid, traversal, stopping, boundary convention
Nuisance parameter ϑ	protocol metadata	fixed, extremized, or probabilistically averaged
Fit window $\mathcal{W}$	index/scale interval	predeclared or selection rule
Weights/covariance	positive weights or matrix	deterministic fit or stochastic model
Estimator	map	WLS, GLS, pairwise range, or alternative
Residual summary	dimensionless	norm and subwindow stability

Field	Type	Required content
Uncertainty statement	semantics	deterministic sensitivity or probabilistic coverage
Defect normalization	typed ratio or not applicable	required before any scalar loss aggregation
Limit claim	theorem or diagnostic	liminf, limsup, limit, or finite-window only

13 Claim audit

Statement	Status	Required hypotheses
Covering counts are monotone in resolution	theorem	fixed metric and cover convention
Box exponents are unit, reference, and base independent	theorem	positive reference and asymptotic limit
Lipschitz observation cannot increase upper box dimension	theorem	matched output scale
Bi-Lipschitz observation preserves box dimensions	theorem	two-sided distance bounds
All reasonable rulers give identical finite traces	false	only asymptotic comparison may hold
Finite log-log slope is a fractal dimension	diagnostic only	becomes a dimension only under a justified limit
Residual slope range is a confidence interval	false	no coverage probability without a stochastic model
Every natural coastline has one universal dimension	not claimed	finite regimes and crossovers are allowed
Koch divider length diverges	exact benchmark	generator-aligned construction scales
Mandelbrot's ruler argument proves an S-model	not claimed	no such implication is used

14 Conclusion

The mathematically stable object is not one measured length. It is the typed protocol trace

$$\left\{ \left(\log_b \frac{\ell_*}{\varepsilon}, Y_{\mathcal{P}}(A; \varepsilon) \right) : \varepsilon \in I \right\},$$

together with its transformation law, asymptotic exponent when it exists, finite-window residuals, and uncertainty semantics. The exponent is robust only under a specified class of observation maps or exponent-equivalent protocols. This is the strict content of the Mandelbrot ruler layer in Geometry of Observation.

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